ADAPI: Arrowhead Data-plane Authorization Policy Interfaces

Extending Arrowhead Authorization to the Data Plane

Gap

ADAPI closes the gap between Arrowhead's control-plane authorization and real-time data-plane enforcement and extends it to cover PKI identity verification across all transport paths.

Arrowhead 5.2 defines how systems register, discover, authenticate, and obtain authorization, but not how authorization is enforced once communication begins, nor what brokers do with client certificates after the TLS handshake. ADAPI extends Arrowhead by projecting authorization policy, including certificate validity and profile level, onto live data flows, allows for the use of external access management solutions.

Key idea

Two typed gRPC interfaces connect Arrowhead authorization and PKI to transport-level enforcement, evolving beyond Arrowhead 5.2.

The ADAPI authorization interfaces unify decision-making and certificate state propagation: It defines the PEP-to-PDP interface to share a single policy store and allows for grant events to be streamed from the CA to all PEPs.

What ADAPI enables

- **Runtime-revocable authorization** across live data flows
- **Cryptographically verified identity** on all transport paths
- **Instant revocation propagation** via gRPC stream
- **Cert-level conditioning** in XACML policy
- **Unified policy authority** for orchestration and data-plane enforcement

Why it matters

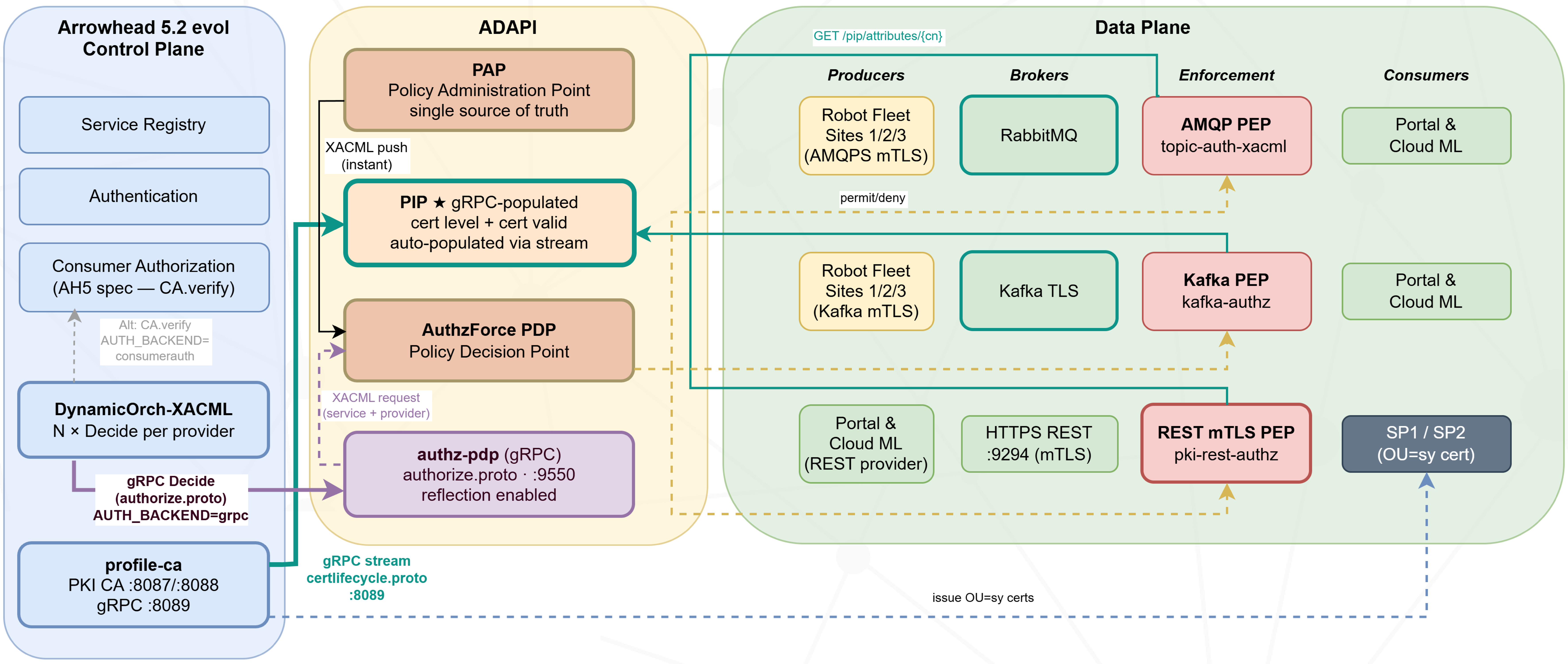
ADAPI lets IoT deployments use different communication protocols for different device classes while being governed by one common authorization and identity model.

- Message-oriented devices can use **AMQP**
- High-throughput actuators can use **Kafka**
- **REST**-capable edge nodes can use request-response flows
- Constrained sensors can use lightweight messaging with **MQTT**
  - *Not implemented in the PoC, but would follow the same principle*

Main contribution

ADAPI creates a closed loop from Arrowhead PKI and control-plane grants to enforceable, identity-verified data-plane policy.

Certificate state and authorization decisions propagate system-wide: certificate state is distributed to all PEPs automatically; decisions are unified under a single interface and policy store. Grants and revocations take effect instantly across transports.



GitHub: https://github.com/ulfbod/Arrowhead-experimental-core-systems
(experiment-13 + core-evol and support systems)

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